

Mining students' behavior in web-based learning programs

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ABSTRACT

There has been a proliferation of web-based learning programs (WBLPs). Unlike traditional computer-based learning programs, WBLPs are used by a population of learners who have diverse background. How different learners access the WBLPs has been investigated by several studies, which indicate that cognitive style is an important factor that influences learners' preferences. However, these studies mainly use statistical methods to analyze learners' preferences. In this paper, we propose to analyze learners' preferences with a data mining technique. Findings in our study show that Field Independent learners frequently use backward/forward buttons and spent less time for navigation. On the other hand, Field Dependent learners often use main menu and have more repeated visiting. Implications for these findings are discussed.

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1. Introduction

Web-based learning programs (WBLPs) change the approaches of delivering instructional materials (Altun & Cakan, 2006) and open new ways of learning for many people (Kinshuk, 2003). One of the differences between WBLPs and traditional computer-based learning programs is that WBLPs provide non-linear learning, which allows students to have freedom of navigation. However, a number of studies, e.g., Chen (2002) and Rogers and Iba (2000), point out that such a learning approach may not be suitable to all learners. Learners may have different background, especially in terms of their knowledge, skills, and needs, so they may exhibit various levels of engagement in course content (Wang, 2007). Thus, individual differences become an essential issue in the development of WBLPs.

Such differences include gender differences (e.g., Roy & Chi, 2003), prior knowledge (e.g., Calisir & Gurel, 2003), and cognitive styles (e.g., Chen & Macredie, 2004). Among them, this study will focus on cognitive styles because they influence the effectiveness of learning and teaching (Miller, 1987). A specific research question is addressed: "how students' cognitive styles affect their learning behavior in a WBLP?". Unlike most of previous studies, which used traditional statistical methods to analyze students' learning behavior, answers to this research question are sought by using a data mining technique to conduct data analyses. This is due to the fact that data mining can uncover hidden relationships, patterns, and interdependencies (Gargano & Ragged, 1999; Hedberg, 1995).

The structure of the paper is structured as follows: Section 2 presents the background, which presents previous works related

to cognitive styles and data mining techniques. Section 3 describes methodology used to conduct an empirical study. Section 4 analyses and discusses the results from the empirical study. Finally, conclusions and future works are presented in Section 5.

2. Background

2.1. Cognitive styles

Web-based learning programs (WBLPs) provide a highly flexible teaching and learning environment for both instructors and students (Pituch & Lee, 2006). Students are able to follow paths through the subject content produced by designers or to develop their own routes (Chen & Macredie, 2004). In addition, learners can read course content through a computer network at any time and from different places (Chang et al., 1998). However, whether learners appreciate such flexibility is greatly affected by individual differences.

Individual differences include a number of psychological factors, one of which is cognitive styles (Wang, Rees, & Liao, 2002). Cognitive style, which is originally proposed by Allport (1937), can be used to describe and explain differences in using strategies among individuals, especially for information representation and information processing (Riding & Rayner, 1998). The most widely examined cognitive style in education is Field Dependence, of which advocates are Witkin and Asch (1948a,b). Riding and Cheema (1991) state that Field Dependence refers to the degree to which a learner's perception or comprehension of information is influenced by the surrounding perceptual or contextual field (Jonassen & Grabowski, 1993). In other words, it describes how well an individual is able to restructure information based on the

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Table 1

Differences between field dependent and field independent learners

Field dependent learners	Field independent learners
They find it difficult to restructure new information and forge links with prior knowledge	They are able to reorganize information to provide a context for prior knowledge
Their personalities show a greater social orientation	They are influenced less by social reinforcement
They experience surroundings in a relatively global fashion, passively conforming to the influence of the prevailing field or context	They experience surroundings analytically, with objects experienced as being discrete from their backgrounds
They demonstrate fewer proportional reasoning skills	They demonstrate greater proportional reasoning skills
They prefer working in groups	They prefer working alone
They struggle with individual elements	They are good with problems that require taking elements out of their whole context
They are externally directed	They are internally directed
They are influenced by salient features	They are individualistic
They accept ideas as presented	They accept ideas strengthened through analysis

Adapted from Witkin et al. (1977), Jonassen and Grabowski (1993), and Morgan (1997).

use of salient cues and field arrangement. The key issue of Field Dependence lies within the differences between Field Dependent and Field Independent learners and Table 1 presents a summary of their differences.

A number of studies investigated how Field Dependent and Field Independent differently reacted to WBLPs. For instance, Lu, Yu, and Liu (2003) investigated students' learning patterns and learning styles in a WebCT course. Seventy six graduate students participated in their study and the group embedded figures test (GEFT) was used to classify the participants' cognitive styles as Field Dependent or Field Independent. Their study found that Field Dependent subjects made more use of the teaching notes and other class resources than Field Independent subjects. Additionally, Graff (2003) examined the relationship between cognitive style and performance in two versions of the system – long-page and short-page versions. Fifty students took part in their study. Riding's Cognitive Style Analysis (1991) was applied to determine an individual's cognitive style. Their results indicated that Field Independent students achieved higher scores in the long-page condition whereas Field Dependent students performed better in the short-page condition. The aforementioned studies and other previous studies (e.g., Chen & Macredie, 2004) provide useful findings but these studies mainly rely on conventional statistical methods to analyze learners' preferences. For example, frequency and percentage were applied in Lu et al. (2003) and MANOVA was used in Graff (2003). Making assumptions is an inevitable step in using such statistical methods. Thus, some relationships may not be able to be discovered if they are neglected in the process of making assumptions. To address this issue, there is a need to use intelligent technologies, one of which is data mining.

2.2. Data mining

Data mining refer to the process of employing one or more machine learning techniques to automatically analyze and extract knowledge from data contained within a database (Roiger & Geatz, 2003). It is a systematic approach for exploring underlying patterns, trend, and relationships hidden in available data (Lee, Chiu, Chou, & Lu, 2006), and it can be used to generate rules to predict correlations (Gargano & Ragged, 1999). One of the advantages of using data mining techniques is that unexpected relationships within a dataset can be discovered (Chang & Chen, 2005; Waite, Wheeler, & Bromfield, 2007).

A number of data mining techniques are available, but they may be divided into two major categories: supervised learning and unsupervised learning. The former refers to assigning objects to predefined categories or classes. (Hastie, Tibshirani, & Friedman, 2001). On the contrary, the latter is concerned with the division of data into groups of similar objects (Chen & Liu, 2004; Nolan, 2002). To determine the appropriateness of using a supervised or an unsupervised learning method, there is a need to consider whether there are known categories (Song, Akande, Idem, & Mahinpey, 2007). If categories are not known, unsupervised learning method ought to be employed. On the other hand, supervised learning method should be adopted. In other words, supervised learning can only be used if the categories are known. This study aims to investigate the relationships between cognitive styles and students' learning behavior so there is a known category, i.e. cognitive styles. Therefore, a supervised learning method may be an appropriate approach for this study.

Classification is a widely used supervised learning technique for extracting meaningful information from a dataset (Baglioni, Furlotti, & Turini, 2005; Roiger & Geatz, 2003). One of popular classification tools is decision tree, which is used to discover rules and relationships by systematically dividing instances found within the dataset (Quinlan, 1993; Wang, Chuang, Hsu, & Keh, 2004). Typically, a decision tree begins with the entire set of data, splits the data into two or more subsets based on the values of features, and then repeatedly splits each subset into smaller subsets until the size of each subset reaches a predefined level. The entire modeling process can be represented in a hierarchical tree structure (Li, 2005). The hierarchical structure can help readers easily understand the relationships hidden in the dataset.

Due to such an useful property, several researchers have used decision trees for classification. In particular, decision trees have been used for classifying users' preferences. Zhu, Greiner, and Haulbl (2003), for instance, used decision trees to construct a recommendation system that helps users find relevant information on the Web. In a similar manner, Liu and Kešelj (2007) proposed a method that uses decision trees to automatically classify users' Web navigation patterns to predict which Web pages are more likely to be accessed next by them. The abovementioned studies indicate that decision trees provide additional benefits for data analysis. We, therefore, attempt to employ such a technique for analyzing the results of an empirical study that investigates the relationships between students' cognitive styles and their learning behavior.

3. Methodology design

3.1. Participants

This study was conducted at an UK university. A total of 65 students participated in this study. All participants had the basic computing and Internet skills necessary to operate a WBLP. At the outset, they were inexperienced in the content domain of the WBLP. Despite the fact that the participants volunteered to take part in the study, they were found to be evenly distributed in terms of cognitive styles. In addition, the participants were an almost equal mix of male and female within each cognitive style group. Fig. 1, illustrates the distribution of the sample of this study.

3.2. Research instruments

Research instrument is a mechanism to help ensure that same information can be obtained from different participants. The research instruments used in this study included a web-based learning program and riding's cognitive style analysis (CSA).

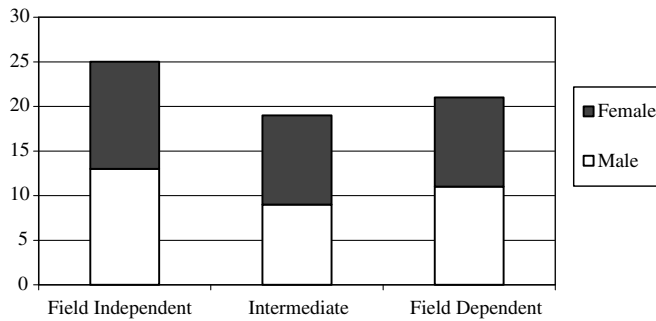


Fig. 1. Distributions of participants.

3.2.1. Web-based learning program

A web-based learning program (WBLP) presents an introduction to HyperText Markup Language (HTML). The interface of the WBLP was divided into three sections, including a title bar, a control panel, and a main body. The title bar was located at the top of the screen, showing the section name being viewed. The navigation panel was located at the left of the screen. It contained several navigation buttons, including main menu, hierarchical map and alphabetical index, with which students can freely navigate in the WBLP. The main body was located at the right of the screen, providing the subject contents and all relevant learning materials. The subject content is divided into three sections: (1) What is HTML? (2) Working with HTML, and (3) relations with SGML and WWW. The key element of the WBLP is located in the second section, where 12 sub-topics are included. Each sub-topic is further split into five parts: (1) overview, (2) detailed techniques, (3) examples, (4) related skills, and (5) references. By doing so, students' preferred content presentation could be undertaken by examining their learning paths.

3.2.2. Cognitive style analysis

The cognitive style investigated in this study was Field Dependence. Group embedded figures test (GEFT) by Witkin, Oltman, Raskin, and Karp (1971) and the cognitive styles analysis (CSA) by Riding (1991) two widely used instruments for the measurement of Field Dependence. Riding's CSA was selected in this study because Field Dependence can be positively measured, instead of inferring from the poor Field Independent capability (Ford & Chen, 2001). Riding's CSA includes two sub-tests, each of which has different purposes. The first sub-test aims at identifying participants' Field Dependent capabilities by requesting participants to determine whether pairs of complex geometrical figures are either the same or different. The second sub-test aims at investigating the disembedding capacity associated with Field Independence by asking participants to identify whether a simple geometrical shape, such as a square or a triangle, is included in a complex geometrical figure.

The CSA measures what Riding and Sadler-Smith (1992) refer to as a wholist/analytic (W/A) dimension, noting that this is equivalent to Field Dependence/Independence. As Witkin et al. (1971) argued, a Field Independent individual is capable of a more analytical cognitive function than a Field Dependent individual, who uses a more global approach. Riding (1991) recommendations are that scores below 1.03 denote Field Dependent individuals; scores of 1.36 and above denote Field Independent individuals; students scoring between 1.03 and 1.35 are classed as Intermediate. In this study, categorizations were based on these recommendations.

3.3. Procedure

The study consisted of three steps. Firstly, the participants were asked to take Riding's CSA test to classify their cognitive

styles. All participants were then given an introduction about how to use this WBLP. Subsequently, they were asked to interact with the WBLP program for 90 min and their interaction was recorded in a log file.

3.4. Data analyses

As described in Background section, decision trees were used to analyze students' learning behavior. Popular decision tree algorithms includes classification & regression trees (C&RT) (Breiman, Friedman, Olshen, & Stone, 1984), chi-squared automatic interaction detection (CHAID) (Kass, 1980), and C4.5 (Quinlan, 1993). In this study, we used the C4.5 because it is a well-known classification algorithm and can generate easily understandable rules (Ding, Ding, & Perrizo, 2002). The main goal of the algorithm is to discover relationships between a given classification of objects and a set of attributes. The output of the algorithm is a classification tree showing how objects may be assigned to the given classes on the basis of values of these attributes (Andrienko & Andrienko, 1999). Tree production has three phases in this algorithm. In Phase I, an initial and large tree is created from the sets of examples according to attribute selection measures. In Phase II, an error-based pruning method is used to simplify the tree by discarding one or more sub-trees and replacing them with leaves or branches (Quinlan, 1993). In Phase III, the classification performance of the decision tree is tested by analyzing the number of correctly and incorrectly classified instances. The number of correct classified instances determines whether the decision tree can be applied to the datasets, or whether further preparation will be necessary.

To further verify the relevance of attributes identified by the decision tree, a series of ANalysis Of VAriance (ANOVA) was adopted to determine the significance of each attribute. A significance level of $p < 0.05$ was adopted for the study.

4. Results and discussion

A decision tree was created to illustrate the learning behavior of each cognitive style group (see Fig. 2). As shown in this figure, the frequency of using the backward/forward button is the most important feature for classifying the cognitive styles and the second most important features are the frequencies of using overviews and those of having repeated visiting. In other words, these three attributes have close relationships with cognitive styles and they should be considered for the development of WBLPs that meet the needs of each cognitive style.

In addition to illustrating the students' learning behavior with a tree structure, the decision tree can also be converted to create rules for the predictions of the students' cognitive styles, i.e., identifying the cognitive styles of new students based on their learning behavior. Table 2 lists 10 rules drawn from the decision tree. As showed in this table, different rules lead to different cognitive styles. A cognitive style could be reached by more than one rule. In this case, three rules are connected with Field Dependent students, four rules are associated with Intermediate students, and three rules are related to Field Independent students. These rules can be applied to replace the CSA or other cognitive style tests and work as criteria for automatic identification of the students' cognitive styles. In other words, the students' cognitive styles can be automatically recognized based on their learning behavior and such automatic mechanism are useful for developing personalized WBLPs that can accommodate students' cognitive styles.

As shown in Fig. 2 and Table 2, the relevant attributes for identifying students' cognitive styles include the use of backward/forward button, overview, example, reference, and main menu, the repeated visiting and time spent for navigation. The ANOVA was

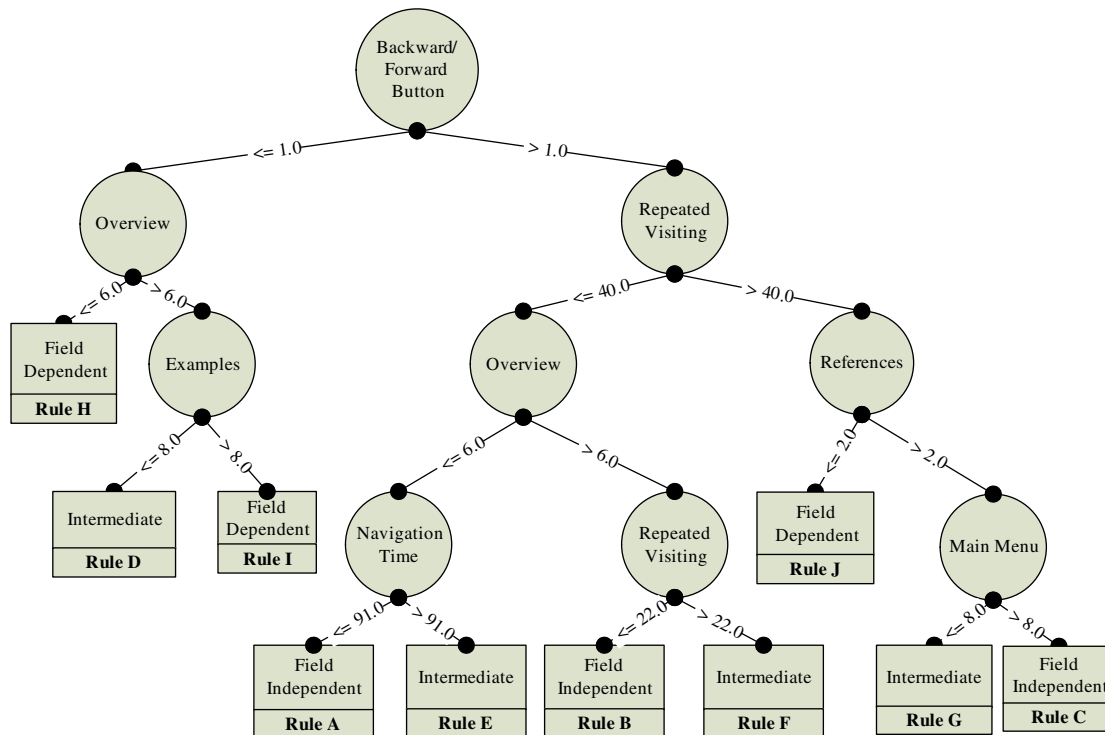


Fig. 2. Decision tree model for CSA vs. navigation tools.

Table 2
Decision rules for different cognitive styles

Cognitive styles	Decision rules
Field independent (rule A)	Learners who have accessed backward/forward button more than once, have repeated visiting more than six times, and spent navigation time for 91 min or less than 91 min
Field independent (rule B)	Learners who have accessed backward/forward button more than once, have repeated visiting 40 times or less than 40 times, select overview more than six times, and have repeated visiting 22 times or less than 22 times
Field independent (rule C)	Learners who have accessed backward/forward button more than once, have repeated visiting more than 40 times, select references more than twice, and use main menu more than eight times
Intermediate (rule D)	Learners who have accessed backward/forward button once or less than once, overview more than six times, and example eight times or less than eight times
Intermediate (rule E)	Learners who have accessed backward/forward button more than once, have repeated visiting 40 times or less than 40 times, select overview six times or less than six times, and spent navigation time for more than 91 min
Intermediate (rule F)	Learners who have accessed backward/forward button more than once, have repeated visiting 40 times or less than 40 times, select overview more than six times, and have repeated visiting more than 22 times
Intermediate (rule G)	Learners who have accessed backward/forward button more than once, have repeated visiting more than 40 times, select references more than twice, and use main menu eight times or less than eight times
Field dependent (rule H)	Learners who have accessed backward/forward button once or less than once, overview six times or less than six times
Field dependent (rule I)	Learners who have accessed backward/forward button once or less than once, overview more than six times, and example more than eight times
Field dependent (rule J)	Learners who have accessed backward/forward button more than once, have repeated visiting more than 40 times, and select references twice or less than twice

used to examine the significance of these attributes and the results indicate that cognitive styles have significant effects on these attri-

butes, apart from the use of references (Table 3). Table 4 presents the mean and standard deviation of the frequency of each attribute for each cognitive style group.

As indicated by the aforementioned results, learners with different cognitive styles demonstrate different learning behavior in using the WBLP. Their differences can be considered as a mirror of the characteristics of different cognitive styles. For example, Field Independent students made less use of overview. It is probably because Field Independent learners tend to take a serialist learning approach that concentrated primarily on procedural details when processing information in a learning context (Pask, 1976, 1979). Therefore, a global picture provided by overview is less interesting to them. Furthermore, they spent less time for navigation. A possible interpretation of this finding is that Field Independent students tend to be more analytical (Ford, Wood, & Walsh 1994) and they are very task oriented (Witkin, Moore, Goodenough, & Cox, 1977). Hence, they only pay attention to particular topics related to their learning, which they, in turn, need less time for navigation. In addition, they favored to use backward/forward buttons. It suggests that Field Independent students feel comfortable to jump freely within a learning environment with backward/forward buttons. In other words, they prefer non-linear navigation. Such a finding is in line with those of Lee, Cheng, Rai, and Depickere (2005) which indicate that Field Independent students have a preference for non-linear learning.

Table 3
Significance of cognitive styles based on ANOVA (df = 2,62)

Attributes	Significance
Backward/forward	$F = 8.857; p < .01$
Overview	$F = 8.496; p < .01$
Repeated visiting	$F = 5.065; p < .01$
Example	$F = 4.892; p < .05$
References	$F = .606; p > .05$
Main menu	$F = 3.876; p < .05$
Navigation time	$F = 7.601; p < .01$

Table 4

Mean frequency (standard deviation) of attributes in each cognitive style

Cognitive styles	Backward/forward	Overview	Repeated visiting	Example	Reference	Main menu	Navigation time
Field independent	4.88 (0.9)	5.12 (0.6)	26.76 (3.4)	4.92 (0.7)	1.72 (0.1)	3.28 (0.3)	79.92 (3.6)
Intermediate	3.00 (0.7)	7.26 (0.5)	32.05 (3.1)	7.16 (0.4)	2.05 (0.9)	4.21 (0.3)	89.21 (0.5)
Field dependent	2.76 (0.9)	5.71 (0.2)	39.14 (0.8)	8.62 (0.4)	1.81 (0.9)	5.81 (0.8)	90.71 (5.7)
All	3.65 (0.2)	5.94 (0.9)	32.31 (0.9)	6.77 (0.2)	1.85 (0.1)	4.37 (0.2)	86.12 (1.2)

In contrast, Field Dependent students often look at examples. Examples are down-to-earth visual material to provide illustrations with practical cases. Therefore, they are beneficial to Field Dependent individuals, who rely more on external frames of reference and operate best where analyses are already provided (Lyons-Lawrence, 1994). Moreover, Field Dependent students frequently use the main menu, which reflects the conceptual structure of the subject content. This might be caused by the fact that Field Dependent students tend to take a passive approach by relying on salient cues (Anastasi, 1988) and have more difficulties in separating the individual parts within a whole (Witkin et al., 1977). With the extra guidance provided by the main menu, Field Dependent students can more easily access meaningful information. Additionally, Field Dependent students have more repeated visiting, which might suggest that they are not very sure where they should go and feel lost in hyperspace. This may be due to the fact that Field Dependent people prefer linear presentation of learning material and have difficulties in non-linear learning. These findings echoed previous research (Witkin et al., 1977); suggesting that Field Dependent students have more difficulties dealing with confusion, complexity, and dimensions which are often present in WBLPs.

On the other hand, diverse decision rules, including rules D, E, F, and G, can be used to identify Intermediate students. It may suggest that they are equally comfortable using each learning path. One possible interpretation is that individuals possessing an Intermediate cognitive style combine the characteristics of both Field Independence and Field Dependence and employ a more versatile strategy for their learning. Versatile students, who have acquired the skill to move back and forth between different learning strategies, are more capable of adapting themselves to suit the learning programs (Ford, 2000). Thus, Intermediate students can be recognized by using a variety of decision rules.

5. Implications for WBLPs development

As the results showed in the previous section, cognitive styles have significant effects on learners' behavior in WBLPs. In particular, they influence the use of backward/forward buttons, the use of overview, and the frequencies of having repeated visiting. The differences in their learning behavior have implications for the development of personalized WBLPs.

Personalization can be delivered by providing adaptivity or adaptability. With respect to the former, learners' cognitive styles can be identified by either monitoring his/her learning patterns with data mining techniques or by obtaining this information from an external survey. According to their cognitive styles, the design of WBLPs could be automatically tailored to match the preferences of each individual. With respect to the latter, users are allowed to modify navigation tools and content presentation by themselves. For example, a checkbox-based form can be applied to identify learners' preferences. Each learner will express his/her preferences by entering a checkmark in a suitable checkbox. The design of the WBLPs will be changed based on his/her choices in the checkbox-based form.

6. Conclusions and future works

Two main conclusions can be drawn from this study. The first is that cognitive style is an important factor that determines students' learning behavior. This conclusion strengthened the findings from previous studies, such as Ford and Chen (2001) and Chen and Macredie (2004). However, it was only one relatively small study. Further work needs to be undertaken with a larger sample to provide additional evidence.

The second is that data mining, more specifically decision trees, is a useful tool for classifying students' cognitive styles. The advantage of using data mining rather than statistical methods is that it is not necessary to make any assumptions. However, this study only used decision trees. Further work can analyse students' learning patterns using other classification methods, such as *k*-nearest neighbor or support vector machines. It would be interesting to see whether similar results would be found by using these classification methods.

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